

Technical reference

The operational reference: exact commands, provider contract checks, framework facts, module boundaries, data limitations, and troubleshooting. How to run the system is covered in the [README](#). How it works - the architecture, the eight agents, the decision safety net, review and resume, payment idempotency, storage, and the web console - is covered with diagrams in [SYSTEM_GUIDE.md](#) ; [DEMO.md](#) is the short business walkthrough.

Database lifecycle

`invoice-agents ui` migrates, seeds, and verifies both databases on start (idempotently). The same steps can be run explicitly - normal case processing never creates or repairs a database:

```
uv run python -m invoice_agents.db migrate --db inventory.db
uv run python -m invoice_agents.db seed --db inventory.db
uv run python -m invoice_agents.db verify --db inventory.db

uv run python -m invoice_agents.db migrate --db workflow.db --kind workflow
uv run python -m invoice_agents.db verify --db workflow.db --kind workflow
```

Verification pins the inventory seed verbatim; it contains only facts from the challenge README:

SKU	Item	Available stock
SKU-WIDGET-A	WidgetA	15
SKU-WIDGET-B	WidgetB	10
SKU-GADGET-X	GadgetX	5
SKU-FAKE-ITEM	FakeItem	0

Alias rows start empty. Fuzzy candidates are suggestions only; an exact item name, persisted approved alias, or recorded human mapping is required to establish a canonical SKU. The schema and storage model are described in [SYSTEM_GUIDE §7](#).

Human review commands

What routes a case to review, the five decisions, and their exact consequences are covered in [SYSTEM_GUIDE §5](#); the policy knobs (`INVOICE_REVIEW_THRESHOLD_*`, `INVOICE_DUE_DATE_TOLERANCE_DAYS`) are listed in [SYSTEM_GUIDE §9](#). Those values are business policy; changing them requires business-owner review.

```
uv run invoice-agents review list
uv run invoice-agents review show REVIEW_ID

uv run invoice-agents review decide REVIEW_ID `
  --reviewer vp@example.com `
  --decision REJECT `
  --reason "Requested quantity is not authorized."

uv run invoice-agents review resume CASE_ID
```

`--decision` accepts `APPROVE`, `REJECT`, `REQUEST_CORRECTION`, `ESTABLISH_MAPPING`, and `SUPERSEDE_REVISION`. Mapping decisions require one or more `--mapping "raw item=SKU"` arguments; revision decisions require `--superseded-case-id`. Reviewer and reason are mandatory. Review packages for PDF sources include rendered page images under `artifacts/reviews/{review_id}/`.

Web console operations

The console's pages and design rules are covered in [SYSTEM_GUIDE §8](#), with the page-by-page plan in [UI_PLAN.md](#). Operational facts: the server binds to localhost only and there is no auth layer - the CLI refuses non-loopback hosts without `--allow-remote-i-understand`. The key value is never rendered, only whether it is present. All assets ship in-repo (no CDN, no Node toolchain). Opt-in browser smokes exist behind `RUN_UI_SMOKE=1` (`uv run playwright install chromium` first).

Provider compatibility contract

The paid live suite proves authentication, the exact model and endpoint, strict typed tools, sequential tool iterations, Pydantic structured output after tool evidence, Swarm handoff, agent-name configuration, stopped-state save/load/human resume, and tool-exception visibility:

```
uv run invoice-agents contract --live
```

Without `--live`, the CLI reports `NOT RUN` and exits nonzero; a skipped contract is never described as a pass. The verified xAI-safe message-name configuration is fixed at `include_name_in_message=False` and `add_name_prefixes=True`, not selected as a fallback.

Tests and quality checks

```
uv run ruff format --check .
uv run ruff check .
uv run mypy src
uv run pytest -m "not live"

# Explicitly opt into paid xAI tests:
$env:RUN_LIVE_XAI="1"
uv run pytest -m live
```

What each suite locks down, per area, is tabulated in [SYSTEM_GUIDE §11](#).

Verification artifacts

`artifacts/verification/` holds the two `workflow_e2e*.db` databases retained from the 2026-08-06 live end-to-end verification of `INV-1001` (clean approval and payment) and `INV-1002` (policy review path), plus archived local workflow databases from completed run phases. The directory is gitignored local evidence.

The Phase 8 full-corpus live reconciliation - all 20 artifacts processed end to end on the remediated system, with the expected-decision script, per-artifact outcomes, exactly five payments, and both run records - is documented in [PHASE8_RECONCILIATION.md](#).

Framework notes

AutoGen 0.7.5 is in maintenance mode and identifies Microsoft Agent Framework as its successor. This prototype remains on AutoGen because that is the selected framework; it does not change frameworks at runtime. The team is a `Swarm` with model-driven handoff tools; `SelectorGroupChat` is intentionally not used.

Module boundaries

Area	Responsibility
<code>src/invoice_agents/models.py</code>	Strict Pydantic contracts and explicit statuses
<code>tools/evidence.py</code>	File hash/metadata, format readers, PDF text/render evidence, provenance-preserving extraction
<code>tools/comparison.py</code>	Read-only SQL, explicit aliases, aggregation, Decimal totals, dates, identity, policy evidence
<code>agents/team.py</code>	Narrow prompts, one composite deterministic tool per specialist plus two read-only critic recheck tools (ADR-001), handoffs, critic loop, approval and payment agent construction
<code>agents/decision_rules.py</code>	Pure final-decision rules; <code>APPROVE</code> requires the critic's agreement or a resolved authorizing human decision
<code>orchestration.py</code>	Preflight, fresh-team streaming, circuit breakers, state save/resume, final status mapping
<code>hitl/</code>	Complete review packages and attributable human decisions
<code>payment/</code>	Approval checks, mock transaction ledger, idempotency, injected failure
<code>db/</code> and <code>migrations/</code>	Explicit versioned database lifecycle and workflow persistence
<code>observability/</code>	Redacted correlated events and local OpenTelemetry spans

Evidence and data limitations

The inventory database authoritatively supports only item existence and available stock. Vendor, purchase order, price, tax, currency/FX, bank-account, and fraud-master reconciliation are explicitly reported as unavailable; agents must not imply those checks occurred. Inventory is an independent per-invoice validation reference - processing does not reserve or decrement stock, so a production batch could overcommit inventory; reservation and shared-commit semantics must be designed before enabling real payment.

PDF extraction uses `pypdf`; `render_pdf_page` uses PyMuPDF when layout evidence is needed. A successful text extraction is not treated as visual proof. xAI vision is not required by the current workflow, so invoice page images are not sent to the provider.

Invoice evidence, agent messages, and results are sensitive local data. `.env`, databases, rendered pages, results, and other runtime artifacts are ignored by Git. Logs recursively redact credential keys and bearer values. AutoGen streamed events are persisted locally; ordinary console logging does not duplicate full prompts. SDK retry attempts are correlated to the active case. AutoGen's pinned OpenAI-compatible client does not expose xAI's Zero Data Retention response header, so each run records that status as `NOT_EXPOSED`

rather than making a ZDR claim. xAI remains the one remote dependency, so business owners must approve its retention terms, serving regions, and any Zero Data Retention requirement before production use.

Troubleshooting

The full failure taxonomy is in [SYSTEM_GUIDE §10](#); common stop reasons and their fixes:

- `DATABASE_MISSING` / `DATABASE_SCHEMA_MISMATCH` : restart `invoice-agents ui` (which migrates and seeds), or run the explicit migrate, seed, and verify commands; case processing will not repair the database.
- `PROVIDER_AUTHENTICATION_FAILED` : verify `XAI_API_KEY` ; there is no alternate model.
- `PROVIDER_RATE_LIMIT_EXHAUSTED` / `PROVIDER_TIMEOUT` : configured bounded retries were exhausted; the case remains failed/incomplete and keeps the provider request ID when available.
- `NEEDS_HUMAN` : inspect the review package, record a reasoned decision, then resume the case.
- `MAX_MESSAGES_EXHAUSTED` : inspect persisted events; the latest prose is not accepted as a result.
- `PAYMENT_FAILED` : inspect the ledger/event error; no success event is synthesized.